

Mohammad Thabet Hassan Alhussin

Cybersecurity & AI-Security Researcher · B.Sc. Cyber Security (CUD, 2026)

3 IEEE Xplore-indexed papers (lead author) · Elsevier Q1 manuscript in preparation · 1st place, national CTF · Open-source research software (PyPI, CI-tested)

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RESEARCH PROFILE

B.Sc. Cyber Security graduate of the Canadian University of Dubai (CGPA 3.46/4.0) seeking a **Research Assistant** position in cybersecurity and applied machine learning. Lead author of three IEEE Xplore-indexed conference papers, with a journal manuscript in preparation for an Elsevier Q1 venue. I study how machine-learning systems can be made **trustworthy for security-critical use**: explainable and calibrated intrusion detection, edge-deployable voice anti-spoofing, reinforcement-learning agents for automated vulnerability discovery, and quantitative risk modelling of software-update channels. Every study is released as reproducible, CI-tested open-source software with model cards and citation metadata.

Research interests: AI security & adversarial ML · explainable AI for intrusion detection · audio deepfake / anti-spoofing · reinforcement learning for automated security testing · secure OTA & update frameworks · LLM-agent security.

EDUCATION

B.Sc. in Cyber Security — Canadian University of Dubai (School of Engineering, Applied Science and Technology) Sep 2022 – May 2026

PUBLICATIONS

1. **M. Thabet**, G. B. Satrya, and A. Kermani, “**SmartInject: Automated SQL Injection Testing Using Deep Q-Learning and LSTM-Based Payload Generation**,” in *Proc. 2025 2nd Int. Conf. on Artificial Intelligence, Metaverse, and Cybersecurity (ICAMAC)*, Dubai, UAE, Oct. 2025, pp. 1–6. DOI: 10.1109/ICAMAC67779.2025.11398683. PUBLISHED · IEEE XPLORE

► Lead author. Hybrid DQN + LSTM agent (OpenAI Gym environment over DVWA) reaching an 80% vulnerability-discovery rate at DVWA “High”, versus 0% for static-fuzzing and DQN-only baselines.

2. **M. Thabet Hassan**, F. Sadek, A. Sami, and A. Kermani Kolankeh, “**Lightweight Voice Deepfake Detection for Smart Mobility Using Multi-Feature Ensemble Learning**,” in *Proc. 2026 IEEE Int. Conf. on Smart Mobility (SM)*, Al Alamein, Egypt, May 2026, pp. 1–2. DOI: 10.1109/SM69703.2026.11614145. PUBLISHED · IEEE XPLORE

► Lead author. Edge-deployable XGBoost ensemble: F1 = 0.95, 95.17% accuracy, <2 MB model, 36.8 ms feature extraction — no GPU or cloud dependency. Code: github.com/MohammadThabetHassan/VoiceGuard.

3. **M. Thabet Hassan**, O. Alraas, and L. A. AbuSerrieh, “**OTA Update-Channel Compromise Risk in Smart Mobility: A Testbed Comparing Governance and Cryptographic Controls**,” in *Proc. 2026 IEEE Int. Conf. on Smart Mobility (SM)*, Al Alamein, Egypt, May 2026, pp. 1–2. DOI: 10.1109/SM69703.2026.11614128. PUBLISHED · IEEE XPLORE

► Lead author. Monte-Carlo testbed (500 runs × 50,000 ECUs). Layered governance reduced median attack blast radius from 27,784 to 46 ECUs (–99.7%) and modelled cost from \$28.7M to \$55K, relative to cryptographic controls alone.

4. **M. Thabet Hassan**, F. Sadek, A. Sami, and M. Khurana, “**XAI-IDS: A Multi-Dataset Explainable AI Framework for Network Intrusion Detection with a Novel XAI Confidence Score**,” manuscript in preparation for *Journal of Network and Computer Applications* (Elsevier, Q1). IN PREPARATION

► Lead author. Ensemble IDS across CIC-IDS-2017, UNSW-NB15 and CSE-CIC-IDS-2018 (16M+ records, 26 attack classes); F1 up to 0.9964; proposed XAI Confidence Score separates correct from incorrect predictions (Cohen’s d = 1.0–1.2), validated with McNemar’s test and paired t-tests. Code + model card: github.com/MohammadThabetHassan/xai-ids.

RESEARCH EXPERIENCE

Undergraduate Researcher — Explainable & Robust Intrusion Detection · xai-ids Oct 2025 – present

- Designed and ran a multi-dataset benchmark (XGBoost, Random Forest, LightGBM, voting ensemble) with SHAP and LIME explanations; engineered the pipeline to process 16M+ flows and to be re-run end-to-end from a single command.
- Proposed and evaluated a new metric (XAI Confidence Score) for flagging low-confidence predictions for analyst review; added adversarial-robustness (FGSM/evasion) evaluation and statistical significance testing.
- Published the artefact with model card, benchmark tables, CITATION.cff, tests and CI, plus a browser-based live inference demo.

Undergraduate Researcher — Voice Deepfake Detection & Vishing Defence · VoiceGuard Sep 2025 – Jun 2026

- Led the lightweight-ensemble study in [2] and extended it into VoiceGuard: XLS-R + AASIST detector at **2.84% EER** on the official ASVspooF 2021 LA evaluation set, Integrated-Gradients explainability, C2PA-signed provenance, NIST SP 800-86-style forensic reports, and a 0.62 MB ONNX edge build.
- Shipped as a documented open-source platform (FastAPI + React, Docker, CI, SECURITY.md, tests) with a public live demo.

Lead Researcher — OTA Update-Channel Risk Testbed (with O. Alraas and Dr. Lobna AbuSerrieh, CUD) Jan 2026 – May 2026

- Co-designed the Monte-Carlo fleet model and the governance-vs-cryptography control comparison in [3]; studied The Update Framework (TUF) reference implementation as the cryptographic baseline.

Lead Researcher — RL-Driven Automated SQL-Injection Testing (with G. B. Satrya and A. Kermani) 2024 – 2025

- Built the Gym environment, DQN agent and LSTM payload generator for [1]; defined the evaluation protocol against static-fuzzing and DQN-only baselines at three DVWA security levels.

OPEN-SOURCE RESEARCH SOFTWARE

- **TrustWeave** — static, deterministic security review of AI-agent / MCP tool configurations (trust-boundary policy engine, 25 adversarial scenario patterns). Published on **PyPI**; Apache-2.0; CI, benchmarks, CITATION.cff. github.com/MohammadThabetHassan/trustweave
- **MIRAGE SOC Lab** — controlled lab for deterministic detection engineering: replayable telemetry, explainable analyst cases mapped to MITRE ATT&CK, and regression tests for a five-rule detection catalogue. TypeScript/React; Playwright e2e. github.com/MohammadThabetHassan/mirage-soc-lab
- **DiskTrace** — read-only forensic recovery for disk images with explicit evidence boundaries and machine-readable export receipts. Rust; CodeQL + release CI. github.com/MohammadThabetHassan/disktrace
- **Real-Time Log Monitoring & Alerting** (C/Bash SIEM-style detector with structured-JSON alerts) · **TokenAbuse-Azure** (Entra ID device-code / Microsoft Graph attack-chain write-up) · **AI Campus-Safety Detection** (YOLOv8m, mAP@0.5 = 0.98, 5.2 ms/frame, with ethics analysis).

PROFESSIONAL EXPERIENCE

Security Intern (Tech Intern, Security Department), part-time — **Madar Alholool Project Management L.L.C (Orbit Solutions)** Dubai · Jun 2025 – Aug 2026

- Designed and set up the internal security infrastructure: logging pipelines, monitoring components, and scripted security checks; refined monitoring rules to reduce false positives using operational data.
- Ran vulnerability assessments and security reviews of internal applications; supported incident-response procedures and integrated security checks into the SDLC with the development team.
- Built and hardened the public web, mail and DNS stack (Alibaba Cloud, AWS EC2): SPF/DKIM/DMARC, TLS, key-only SSH, firewall rules, patching and access reviews.

AWARDS & COMPETITIONS

- **1st place** — **Exploit3rs National CTF** (Zayed Sector & CyberE71), 2025: first across web, network and binary-exploitation tracks.
- **1st place & “Best Implementation Champion”** — **Dubai AI System Competition** (Alibaba Cloud × RTA Hackathon), 2024: ML smart-map “SmartTranspo” for Dubai taxi drivers; honoured at GITEX Global.
- **1st place** — **CUD First Coding Competition** (2023) · **2nd place** — **CUD Engineering, Software Category** (2024) · **9th** — **Sharjah Police CTF** (2025).

RESEARCH & TECHNICAL SKILLS

Methods	Experimental design & ablation studies · statistical validation (McNemar’s test, paired t-test, Cohen’s d, EER/DET curves) · Monte-Carlo simulation · adversarial robustness evaluation (FGSM, evasion) · explainability (SHAP, LIME, Integrated Gradients) · model calibration & confidence scoring
ML / Data	PyTorch, scikit-learn, XGBoost, LightGBM, Stable-Baselines3 (DQN, PPO), wav2vec2 / XLS-R, AASIST, YOLOv8, ONNX Runtime, captum, SMOTE, pandas/NumPy
Datasets & benchmarks	ASVspooof 2021 LA · CIC-IDS-2017 · UNSW-NB15 · CSE-CIC-IDS-2018 · DVWA / OWASP Juice Shop test environments · Roboflow (custom 10k-image dataset)
Security	Web/network/wireless penetration testing (OWASP Top 10, OWASP LLM Top 10), MITRE ATT&CK & ATLAS, detection engineering & SIEM/log analysis, cloud & identity attacks (Entra ID, OAuth device-code), digital forensics (NIST SP 800-86), Kali Linux, Burp Suite, Metasploit, Wireshark, Nmap
Engineering	Python, C/C++, Rust, Java, TypeScript/JavaScript, SQL, Bash · FastAPI, React, Django · Docker, GitHub Actions CI, CodeQL · AWS, Alibaba Cloud, Azure/Entra · PostgreSQL/MySQL
Scholarly	LaTeX & IEEE manuscript preparation · literature review · reproducible research packaging (model cards, CITATION.cff, benchmarks, Makefiles) · conference presentation (IEEE SM 2026)

CERTIFICATIONS & TRAINING

Security: Penetration Testing / Ethical Hacker (Cisco NetAcad, 2026) · Ethical Hacking Essentials (EC-Council) · Google Cybersecurity Professional · OPSWAT ICIP · Nomu AI Ghurair Cybersecurity Bootcamp	AI & CS: Google AI Professional Certificate (2026) · Introduction to Modern AI (Cisco NetAcad, 2026) · Data Structures (UC San Diego) · Java & OOP (UPenn) · Databases (Meta) · GitHub Foundations · AWS Educate Cloud 101
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Workshops attended: Exploit3rs “Router Hacking” and “Network Pentesting 101” (hands-on labs, 2024). Odoo Summer Bootcamp 2026.

LANGUAGES & REFEREES

Languages: Arabic — native · English — B2 upper-intermediate (TOEFL iBT 80/120, May 2026); all publications and code written in English.	Referees (letters on file): Dr. Lobna Abuserrieh, School of Engineering, Applied Science and Technology, CUD (Lobna.abuserrieh@tud.ac.ae) · Prof. Firuz Kamalov, Department of Computational Sciences, CUD.
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